Mutual Evaluation Between Task Difficulties and LLM Capabilities

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Abstract

Evaluating large language models (LLMs) nowadays relies on two disparate sources of supervision: static benchmarks with ground-truth answers and open-ended arena comparisons based on human preferences. These settings provide complementary but unaligned signals, and existing evaluation methods typically treat them separately, overlooking item-level difficulty and the varying informativeness of questions. We propose a unified, model-centric framework that jointly models item difficulty and model ability across both static and open-ended data using an IRT-style formulation. Our approach produces a shared difficulty scale for heterogeneous evaluation items, enables difficulty-aware model comparison, and facilitates principled dataset construction. Applying the method to five math benchmarks and filtered arena logs, we obtain consistent model rankings across evaluation sources and recover correctness and preference labels with high accuracy. The resulting item and model parameters offer interpretable diagnostics and highlight structural limitations of current benchmarks. Our framework provides a novel foundation for more scalable, informative, and adaptive LLM evaluation.

1. Introduction

The ultimate goal of LLM evaluation is to reliably estimate and differentiate model capabilities across a range of tasks. A central challenge lies in determining the most appropriate sets of questions to assess these capabilities. Broadly, evaluation tasks can be grouped into two categories: (1) static benchmarks with provided ground-truth answers, which are typically well-structured (Hendrycks et al., 2020), and (2) open-ended tasks proposed by the users without explicit ground truth, which better reflect realistic use cases but rely on human feedback for evaluation, as exemplified by Chatbot Arena (Chiang et al., 2024). Currently, the evaluation procedures for these two categories remain largely disjoint, each with its own limitations. Static benchmarks often suffer from saturation and data contamination issues (Ott et al., 2022; Magar & Schwartz, 2022), while crowd-sourced open-ended evaluations require substantial post-processing and quality control to ensure reliability and consistency (Li et al., 2024). Traditionally, evaluation data in both settings are chosen based on human expertise and user needs; in this project, we augment this process with model-centric signals that capture item difficulty and their ability to distinguish model capabilities.

Our main goal is to develop a unified framework for ranking the model-centric difficulty of evaluation data in both static and open-ended arena settings, and to jointly estimate model strengths while incorporating such item-level information. From the data perspective, this provides a shared difficulty scale across heterogeneous sources, enabling more coherent comparisons, and offering item-level difficulty labels that can guide future dataset design (e.g., the construction of harder or better-balanced evaluation datasets). From the model perspective, incorporating item-level signals addresses important limitations of existing metrics: accuracy for static benchmarks treats all questions as equally informative, and arena's Bradley–Terry model (Bradley & Terry, 1952) aggregates win rates without accounting for variations across items. A difficulty-aware framework instead models the interaction between a model's capability and the intrinsic complexity of each question, yielding more reliable and interpretable estimates of comparative model performance.

To combine the difficulty measures of static and open-ended questions into a unified scale, we draw inspiration from classical formulations in Item Response Theory (IRT) (Ackerman et al., 2003; Lord & Novick, 2008; Chen et al., 2025). A key advantage of IRT is that both item difficulty and model ability are placed on the same latent continuum, allowing users to reason about evaluation tasks and model capabilities in a directly comparable way. Methodologically, we extend this framework in two directions. First, for static benchmarks, we employ a 2-parameter logistic (2PL) model that jointly estimates model ability and item difficulty/discrimination, providing a richer signal than sim-

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

ple accuracy. Second, for open-ended Arena data, where the feedback consists of pairwise preferences rather than correctness labels, we introduce a pairwise IRT formulation that expresses the likelihood of one model outperforming another in terms of the same latent variables governing static items. This generalizes the Bradley–Terry model while preserving conceptual compatibility with IRT. Finally, we develop a unified estimation procedure that learns all parameters jointly, ensuring that static and open-ended questions contribute to a shared difficulty scale and enabling consistent, difficulty-aware comparisons across models and data sources.

We apply our proposed approach to a combined dataset of math-related benchmark questions and filtered arena logs, and evaluate 10 different LLMs across these question items. The resulting rankings of question difficulties and model abilities are shown in Figure 1. We compare model rankings derived from static data alone, arena data alone, and the combined dataset, as well as the live public math-focused leaderboard released by LMArena. Across all sources, we observe strong correlations in the final rankings, as shown in Table 1. Using our estimated model strengths and question difficulties, we further infer the original correctness labels in the static setting and the pairwise preference labels in the Arena setting, achieving accuracies of 80.0% and 62.4%, respectively. These results collectively support the soundness and robustness of our proposed framework.

Overall, this unified framework offers a principled foundation for difficulty-aware LLM evaluation. By jointly modeling item hardness and model ability, it supports more reliable model comparison, more informed dataset construction, and targeted identification of capability gaps. The estimated difficulty and ability scores can guide benchmark curation, adaptive testing, leaderboard design, and evaluation of future models, providing a scalable way to align evaluation practices with the rapidly evolving landscape of LLM capabilities.

2. Related Work

2.1. Static and Open-ended Tasks for LLM Evaluation

Static benchmarks have long served as the primary means of evaluating the cognitive performance of LLMs, covering diverse tasks such as natural language inference and semantic understanding (Bowman et al., 2015; Williams et al., 2018; Wang et al., 2018), generic question answering (Rajpurkar et al., 2016; Yang et al., 2018; Kwiatkowski et al., 2019; Liang et al., 2022), and more reasoning-heavy domains like mathematics (Hendrycks et al., 2020; Cobbe et al., 2021; Hendrycks et al., 2021).

More recently, open-ended human preference evaluations such as Chatbot Arena (Chiang et al., 2024) and MT-Bench (Zheng et al., 2023) have become popular for assessing holistic model quality. While these tasks better capture user-facing performance, they require substantial quality control and significant human labors. Several recent benchmarking efforts have started to integrate both static and open-ended tasks (Li et al., 2024; Ni et al., 2024) in order to preserve merits from both sides, though difficulty-aware modeling of items remains unexplored. Our work builds on this line by providing a unified, item-level framework that applies across both benchmark and open-ended contexts.

2.2. Item Response Theory (IRT)

Traditional IRT has been widely used in educational testing and psychometrics (Embretson & Reise, 2013), and several works have explored its application to NLP and ML evaluation. Prior studies have applied IRT to analyze benchmark difficulty (Lalor et al., 2019) and evaluate classifier robustness (Martínez-Plumed et al., 2019), while more recent efforts focus on improving data efficiency (Liao et al., 2025) and reducing reliance on human labels during evaluation (Robertson, 2025). However, existing approaches typically operate only on static datasets or treat preference-based evaluations separately. In contrast, our work provides the first unified, model-centric IRT-style framework that jointly handles static correctness data and open-ended human preference data to estimate item difficulty and model ability on a shared scale.

3. Methods

In this section, we present our unified difficulty-aware evaluation framework, which integrates correctness-labeled static benchmarks with preference-based open-ended arena data. Our approach combines the classical 2-parameter logistic Item Response Theory (2PL-IRT) model with a novel pairwise formulation (Pairwise IRT) designed to operate on human preference data. Together, these components enable a shared latent scale for model ability and question difficulty across heterogeneous evaluation sources. The pseudocode for our algorithm is provided in Algorithm 1.

3.1. Standard 2PL-IRT for Static Tasks

For static benchmarks with ground-truth answers, we model the probability that a model m_i answers a question q correctly using the 2-parameter logistic (2PL) model:

$$P(\text{correct}_{i,q}) = \sigma\big(a_q(\theta_i - b_q)\big) \tag{1}$$

where θ_i is the latent ability of model m_i, b_q is the item's difficulty, k_q is an unconstrained discrimination parameter with $a_q = \exp(k_q) > 0$ as its positive-valued discrimination coefficient, and $\sigma(x) = 1/(1 + e^{-x})$.

Let $\mathcal{D}_{\text{static}} = \{(i, q, y_{iq})\}$ denote the set of static responses,

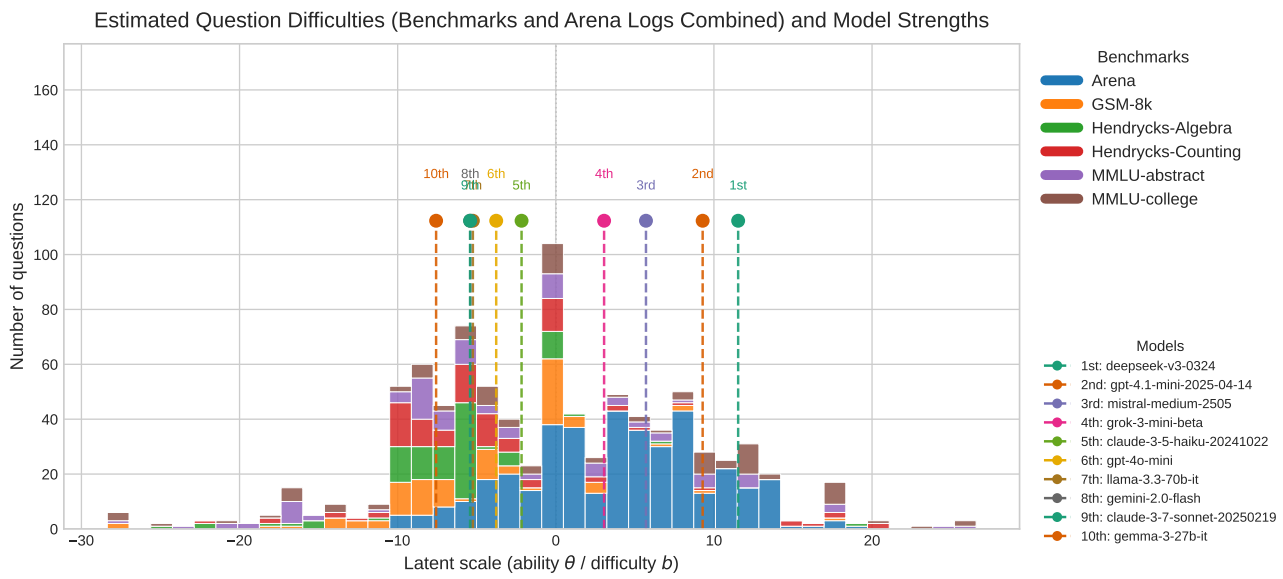

Figure 1. Rankings of model abilities and model-centric question difficulties given by our algorithm.

where $y_{iq} = 1$ if model m_i answers question q correctly and 0 otherwise. The 2PL static-task log-likelihood is:

$$\mathcal{L}_{\text{static}}(\theta, b, k) = \sum_{(i,q)\in\mathcal{D}_{\text{static}}} \Big[y_{iq} \log \sigma\big(a_q(\theta_i - b_q)\big) + (1 - y_{iq}) \log\big(1 - \sigma(a_q(\theta_i - b_q))\big) \Big],$$

where $a_q = \exp(k_q)$.

Compared to naive accuracy-based aggregation, this 2PL formulation accounts for heterogeneous item informativeness: questions with high a_q more sharply differentiate between models of differing abilities, while low-a_q questions act as weak signals. Model parameters are learned by maximizing the Bernoulli log-likelihood over correctness labels, subject to the identifiability constraint $\sum_i \theta_i = 0$.

3.2. Proposed Pairwise IRT for Open-ended Tasks

Arena-style evaluations do not provide correctness labels; instead, they contain pairwise human preferences between model responses. To place these comparisons on the same latent scale as static benchmarks, we introduce a pairwise IRT formulation that links preferences to underlying model abilities and item difficulty.

For a comparison between two models m_i and m_j on question q, we define:

$$P(m_i \succ m_j \mid q) = \sigma\Big(\exp(k_q)\big[\sigma(\theta_i - b_q) - \sigma(\theta_j - b_q)\big]\Big). \tag{2}$$

The inner sigmoids represent each model's latent probability of producing a satisfactory answer given q, while the outer sigmoid converts their difference into a preference probability. This generalizes the Bradley-Terry model by embedding comparisons within the same nonlinear IRT structure used for static data.

Preference Data with Clear Winners For arena comparisons with a decisive winner, we directly use the pairwise IRT likelihood. Let $\mathcal{D}_{\text{win}} = \{(i, j, q, y_{ijq})\}$ denote the set of comparisons in which annotators prefer one model over the other, where $y_{ijq} = 1$ indicates $m_i \succ m_j$ and 0 otherwise. The win–loss term of the arena likelihood is

$$\mathcal{L}_{\text{win}}(\theta, b, k) = \sum_{(i,j,q)\in\mathcal{D}_{\text{win}}} \Big[y_{ijq} \log P(m_i \succ m_j \mid q) + (1 - y_{ijq}) \log\big(1 - P(m_i \succ m_j \mid q)\big) \Big].$$

Preference Data with Ties and Low-Quality Responses Arena annotations also include *tie* and *both_bad* labels. Tie annotations are known to be noisy and inconsistent, and we therefore **exclude tie-labeled comparisons** from the likelihood.

The *both_bad* category is more informative: it indicates that both models produce low-quality responses to question q. Let $\mathcal{D}_{\text{bb}}$ denote these pairs. We encourage low latent success probabilities for both models via the log-likelihood term

$$\mathcal{L}_{\text{bb}}(\theta, b) = \sum_{(i,j,q)\in\mathcal{D}_{\text{bb}}} \Big[\log\big(1 - \pi_{i,q}\big) + \log\big(1 - \pi_{j,q}\big) \Big],$$

where $\pi_{i,q} = \sigma(\theta_i - b_q)$.

Arena Likelihood Combining these components, the arena log-likelihood used in our model is

$$\mathcal{L}_{\text{arena}}(\theta, b, k) = \mathcal{L}_{\text{win}}(\theta, b, k) + \lambda_{\text{bb}}\, \mathcal{L}_{\text{bb}}(\theta, b),$$

with $\lambda_{\text{bb}} > 0$ controlling the strength of the *both_bad* signal.

3.3. Joint Estimation Across Static and Arena Data

With the 2PL likelihood for static data and the pairwise IRT likelihood for arena comparisons, we jointly estimate all parameters to place static and open-ended tasks on a shared latent scale. Let $\mathcal{D}_{\text{static}}$ denote correctness-labeled items and $\mathcal{D}_{\text{arena}} = \mathcal{D}_{\text{win}} \cup \mathcal{D}_{\text{bb}}$ the subset of Arena pairs used in training. The overall log-likelihood is

$$\mathcal{L}_{\text{joint}}(\theta, b, k) = \lambda_{\text{static}}\, \mathcal{L}_{\text{static}}(\theta, b, k; \mathcal{D}_{\text{static}}) + \lambda_{\text{arena}}\, \mathcal{L}_{\text{arena}}(\theta, b, k; \mathcal{D}_{\text{arena}}),$$

and we add an ℓ_2 regularization penalty $\mathcal{R}(\theta, b, k) = \alpha\big(\|\theta\|_2^2 + \|b\|_2^2 + \|k\|_2^2\big)$ for numerical stability (with $\alpha = 10^{-4}$ in our experiments). We maximize $\mathcal{L}_{\text{joint}} - \mathcal{R}$ under the identifiability constraint $\sum_i \theta_i = 0$. Together, these components yield a unified, difficulty-aware evaluation framework applicable to heterogeneous LLM assessment settings.

4. Experiments

We evaluate our framework on a combined dataset consisting of static math benchmarks and open-ended math-related questions extracted from arena logs. While this work currently focuses on the mathematics domain for clarity and reproducibility, the methodology is generalizable and can be applied to any collection of static benchmarks or open-ended tasks, regardless of skill domain.

4.1. Data and Models

Static Data We select five widely used math benchmarks: GSM-8K (Cobbe et al., 2021), two subsets of MATH (algebra and counting_and_probability) (Hendrycks et al., 2021), and two subsets of MMLU (college algebra and abstract algebra) (Hendrycks et al., 2020). These benchmarks cover a range of mathematical domains with varying levels of complexity and abstraction from the human perspective. To maintain a balanced mix of question sources, we subsample the larger benchmarks, resulting in a total of 500 benchmark questions.

Arena Data We filter the released Arena logs (available at `lmarena-ai/arena-human-preference-140k`) to identify user prompts completely within the math domain using an external LLM, and we manually review the filtered questions to ensure alignment with our criteria. Following this process, we select 400 math-related Arena questions for difficulty ranking.

Evaluated Models We evaluate 10 lightweight yet competitive models selected from the Arena leaderboard, covering major LLM families in the market while keeping API costs manageable. The full list of models is provided in Table 1.

4.2. Obtaining Correctness and Preference Labels from LLM Judges

To ensure fair comparisons, we do not provide answer options during evaluation, even for math benchmarks originally presented in multiple-choice format. For static benchmark questions, we determine correctness by comparing the model's generated answer with the ground-truth solution using an LLM judge. This judge is capable of reliably assessing whether a free-form model response matches the correct answer.

For open-ended questions, a major challenge lies in the sparsity and variability of real human feedback in the arena logs: any given question is evaluated on only one model pair, and the quality of human annotations can vary across different annotators and over time. To avoid these issues and obtain a fully controlled and uniformly calibrated set of supervision signals, we **reconstruct all arena preference labels using the same LLM judge**. For each question–pair (m_i, m_j), we present both model responses to the judge and obtain one of four structured labels: *model_a*, *model_b*, *tie*, or *both_bad*. This process produces dense and relatively consistent preference data that are completely independent of the original human arena votes, allowing us to train the pairwise IRT model on a unified labeling scheme.

We further validate the reliability of the LLM-judge labels through manual review of randomly sampled model pairs. When the two model responses differ in correctness, or when both are incorrect, the judge's decisions closely match human judgments. Even when both answers are correct, its non-tie preferences generally align with human impressions of solution quality. These results indicate that the judge provides a stable, coherent supervision signal in settings where Arena annotations are sparse or inconsistent. As noted earlier, we exclude *tie* labels due to their noise but retain *both_bad* labels, which consistently identify cases where neither model solves the item.

4.3. Results

For all experiments, we optimize the joint IRT objective using the Adam optimizer with a learning rate of 0.02 and train for up to 5000 iterations. Model abilities, item difficulties, and discrimination parameters are initialized to zero, and we apply a small ℓ_2 regularization term (10^{-4}) to stabilize training and prevent overly sharp discrimination values. To ensure identifiability, we re-center model abilities at each iteration by enforcing a zero-mean constraint and shifting

Algorithm 1 Joint Estimation of Model Abilities and Task Difficulties

Input: static data $\mathcal{D}_{\text{static}}$, arena data $\mathcal{D}_{\text{arena}}$, learning rate η, number of iterations T
Initialize: model abilities $\theta_i \leftarrow 0$, item difficulties $b_q \leftarrow 0$, discriminations $k_q \leftarrow 0$
Define $a_q = \exp(k_q)$ for each question q
for $t = 1$ **to** T **do**
  for each (i, q) in $\mathcal{D}_{\text{static}}$ **do**
    $\pi_{i,q} = \sigma(a_q(\theta_i - b_q))$
    $\ell_{\text{static}} \mathrel{+}= y_{iq} \log \pi_{i,q} + (1 - y_{iq}) \log(1 - \pi_{i,q})$
  end for
  for each (i, j, q, y_{ijq}) in decisive arena pairs **do**
    $\pi_{i,q} = \sigma(\theta_i - b_q)$
    $\pi_{j,q} = \sigma(\theta_j - b_q)$
    $p_{ijq} = \sigma(a_q(\pi_{i,q} - \pi_{j,q}))$
    $\ell_{\text{win}} \mathrel{+}= y_{ijq} \log p_{ijq} + (1 - y_{ijq}) \log(1 - p_{ijq})$
  end for
  for each (i, j, q) in both_bad pairs **do**
    $\pi_{i,q} = \sigma(\theta_i - b_q)$
    $\pi_{j,q} = \sigma(\theta_j - b_q)$
    $\ell_{\text{bb}} \mathrel{+}= \log(1 - \pi_{i,q}) + \log(1 - \pi_{j,q})$
  end for
  $\mathcal{L} = \lambda_{\text{static}}\ell_{\text{static}} + \lambda_{\text{arena}}\ell_{\text{win}} + \lambda_{\text{bb}}\ell_{\text{bb}} - \alpha \left(\|\theta\|_2^2 + \|b\|_2^2 + \|k\|_2^2\right)$
  Update θ, b, k using Adam step with learning rate η
  Shift θ to have zero mean and shift b accordingly
end for
Output: Estimated parameters θ_i, b_q, k_q

item difficulties accordingly. We set $\lambda_{\text{static}} = \lambda_{\text{arena}} = 1.0$ to assign equal weight to static and arena questions, and choose $\lambda_{\text{bb}} = 0.3$ for the *both_bad* signal. These optimization choices consistently lead to stable convergence across datasets and model pools.

The rankings given by our algorithm are illustrated in Figure 1. The model abilities span a wide yet coherent range on the latent scale, and their placement relative to the item difficulty distribution indicates which regions of the question pool effectively separate model performance and which regions are already saturated.

4.4. Validation of Ranking Qualities

4.4.1. Ranking of Model Capabilities

To cross-validate the soundness of our overall model ranking, Table 1 compares it with rankings derived from static data alone (using accuracy, 2PL-IRT, and simulated pairwise IRT), rankings obtained from arena data alone, and the math-focused public leaderboard released by LMArena (available at `https://huggingface.co/spaces/lmarena-ai/lmarena-leaderboard`). The simulated pairwise IRT from static data refers to running the pairwise IRT algorithm on preference labels that are synthetically constructed from correctness labels, rather than using correctness directly. This allows us to evaluate whether the pairwise formulation can recover reasonable model rankings when only relative comparisons are available. Here we summarize several meaningful findings:

1. With only static data, the rankings produced by simple accuracy, 2PL-IRT, and simulated pairwise IRT are highly consistent. This indicates that 2PL yields essentially the same model ordering as accuracy while additionally providing item-level difficulty and discrimination estimates. It also shows that even in the absence of absolute correctness labels, well-constructed pairwise preference labels are sufficient for recovering high-quality model rankings through our proposed pairwise IRT formulation.

2. The model ranking obtained from pairwise IRT on open-ended Arena data is broadly consistent with the public LMArena leaderboard. This suggests that our LLM-judge based preference labels, when combined with the pairwise IRT framework, produce rankings that closely reflect those derived from real human feedback and larger-scale Arena interactions. The main discrepancy arises with *Gemini-2.0-flash*, which receives the lowest rank in our estimation but appears higher in the public leaderboard. Upon inspecting its responses, we find that the model often provides only a final answer with minimal explanation, even when the prompt does not explicitly ask for conciseness. This tendency makes its responses less likely to be preferred by the LLM judge, especially in cases where both models

produce correct answers but one offers a more detailed solution. It is also plausible that human evaluators may appreciate direct, concise answers in math contexts, which could explain part of the difference between our rankings and those observed in the public Arena leaderboard.

3. The overall ranking based on the combined dataset is correlated with both the static-only and arena-only rankings. In particular, the highest-ranked models largely agree with the ordering derived from arena comparisons, while the lower-ranked models align more closely with the static-only ranking. This pattern reflects the complementary strengths of the two data sources: arena preferences offer finer discrimination among strong models, whereas static correctness labels provide clearer separation among weaker ones. This trend can also be observed from Figure 1, where the main cluster of arena questions appears to be harder than the majority of static benchmark questions.

4.4.2. Predicting Correctness and Preference Labels from Estimated Rankings

We further evaluate the quality of our learned rankings through a self-supervised prediction task. Given the estimated model abilities and question difficulties, we use the corresponding IRT probability models to predict the original labels: correctness in the static setting and pairwise preferences in the arena setting.

For static questions, prediction follows Equation 1:

$$\hat{y}_{iq}^{\text{static}} = \begin{cases} 1, & \sigma\big(a_q(\theta_i - b_q)\big) \geq 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

For arena comparisons, we reuse Equation 2:

$$P(m_i \succ m_j \mid q) = \sigma\Big(a_q\left[\sigma(\theta_i - b_q) - \sigma(\theta_j - b_q)\right]\Big),$$

and infer decisive win/loss outcomes according to

$$\hat{y}_{ijq}^{\text{win/loss}} = \begin{cases} \text{model_1}, & P(m_i \succ m_j \mid q) \geq 0.5, \\ \text{model_2}, & P(m_i \succ m_j \mid q) < 0.5. \end{cases}$$

The *both_bad* category is handled separately using the latent success probabilities

$$\pi_{i,q} = \sigma(\theta_i - b_q), \qquad \pi_{j,q} = \sigma(\theta_j - b_q),$$

and predicting

$$\hat{y}_{ijq}^{\text{both_bad}} = \text{both_bad} \quad \text{if} \quad \pi_{i,q} < \tau \text{ and } \pi_{j,q} < \tau.$$

In our experiments, we use a fixed threshold of $\tau = 0.001$, chosen to match the low-success regime implied by the training objective.

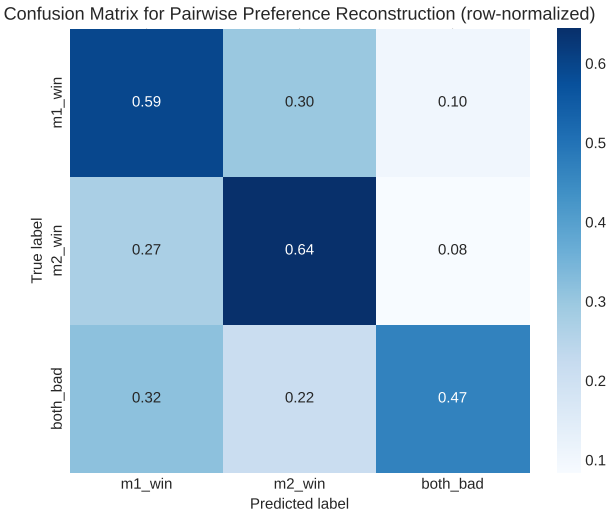

Figure 2. Confusion matrix of using our rankings to predict pairwise preference labels for open-ended arena data. The fractions are normalized across each row.

Comparing these predictions with the ground-truth labels provides an additional check on how well the inferred latent parameters capture the underlying response patterns. Empirically, our estimated rankings achieve an accuracy of 80.0% in predicting static correctness labels and an accuracy of 62.4% in the three-class Arena prediction task (*model_a*, *model_b*, *both_bad*). The corresponding confusion matrix for the Arena setting is shown in Figure 2. Although our preference labels are generated by an LLM judge rather than human annotators, the resulting accuracy is comparable to what fine-tuned reward models typically achieve on math-focused or reasoning-intensive arena-style preference prediction tasks.

5. Discussion

Our work develops a unified, difficulty-aware evaluation framework that integrates correctness-labeled static benchmarks with preference-based open-ended arena data. The empirical analyses presented earlier demonstrate that a shared latent scale for item difficulty and model ability can be reliably inferred across heterogeneous evaluation sources. Here, we discuss broader implications, limitations, and potential future directions.

Broader Implications A key contribution of our framework is bridging the traditional divide between benchmark accuracy and arena win-rate based evaluation. By placing correctness and preference data into a shared probabilistic model, we obtain rankings that align with static-only and arena-only baselines while also producing interpretable

Model	Accuracy	2PL	Pairwise IRT (Static)	Pairwise IRT (Open-ended)	2PL+Pairwise IRT (Overall)	Public Leaderboard
Deepseek-v3-0324	3	3	3	1	1	#41 - #86
GPT-4.1-mini-2025-04-14	2	2	2	2	2	#73 - #96
Mistral-medium-2505	4	4	4	3	3	#64 - #94
Grok-3-mini-beta	1	1	1	5	4	#44 - #90
Claude-3.5-haiku-20241022	5	5	5	8	5	#136 - #158
GPT-4o-mini	6	7	7	6	6	#116 - #148
Llama-3.3-70b-it	8	6	6	9	7	#120 - #152
Gemini-2.0-flash	7	8	8	10	8	#70 - #94
Claude-3.7-sonnet-20250219	9	9	9	4	9	#89 - #110
Gemma-3-27b-it	10	10	10	7	10	#90 - #116

Table 1. Model ranking comparisons using different datasets and ranking algorithms, as well as public leaderboard from LMArena (for which an interval is provided for each model).

item difficulty estimates. This unified view clarifies the complementary strengths of each data source: static benchmarks provide clean signals for distinguishing weaker models, while arena preferences help separate stronger models on harder items. Beyond model ranking, the estimated difficulty parameters reveal structural patterns in existing evaluation data. For example, difficulty distributions highlight saturation in many static benchmarks and motivate the construction of harder, more discriminative items. Meanwhile, difficulty scores also enable adaptive evaluation strategies, such as selecting a small but informative subset of questions or identifying underrepresented difficulty regions that may bias model comparisons.

Limitations and Future Directions Our study focuses on math tasks as a controlled testbed, but extending the framework to less structured settings may require richer modeling assumptions. Additionally, the use of an LLM judge introduces potential biases tied to the judge's reasoning style and training distribution, and this could motivates future work to explore improved calibration, multiple judges, or even partial human supervision. Finally, difficulty-aware evaluation naturally suggests adaptive testing and active benchmark construction, both of which offer promising directions for building more efficient and informative LLM evaluation pipelines.

6. Conclusion

In this work, we present a unified, model-centric framework for difficulty-aware evaluation that integrates static correctness data and open-ended arena preferences into a common probabilistic structure. Our approach jointly estimates item difficulty and model ability, producing coherent rankings that align with existing benchmarks and public leaderboards while offering new diagnostic insights. Experiments on math-related benchmarks and arena logs demonstrate that difficulty-aware modeling improves interpretability, supports principled dataset construction, and enables more reliable comparisons among models of varying strengths. As LLMs continue to advance, we believe such unified and adaptive evaluation frameworks will play an increasingly important role in ensuring that evaluation practices remain informative, efficient, and aligned with real-world model behavior.

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